ThetaSwap: Adverse Competition Oracle

The Risk Nobody Hedges

Claim: Even if impermanent loss is zero, passive LPs carry residual risk from the *competition structure* of fee revenue [Capponi et al., 2024, Aquilina et al., 2024].

$$\Pi_{\mathrm{LP}} = \frac{\text{Fee Revenue}}{1 + \mathrm{LVC}} - \mathrm{LVR}$$

LVR = Loss Versus Rebalancing (adverse selection) **LVC** = Loss Versus Competition (adverse competition, $\approx \Delta^+$; isomorphic to FLAIR [Milionis et al., 2024])

Risk factor		**Hedging instrument**
IL / LVR	$\longrightarrow$	Variance perpetuals (long volatility) [Fukasawa et al., 2023]; deployed by Panoptic [Kristensen, 2024]
LVC	$\longrightarrow$	**??**

If passive LPs are economically *short* fee concentration, then an on-chain index that tracks concentration in real time enables instruments whose payoff is tied to it — letting LPs hedge or insure against adverse competition.

Research Question

Is there empirical demand for a hedging instrument against adverse competition? If so, how much are LPs willing to pay?

Data

ETH/USDC 30bps pool (Uniswap V3 mainnet), Dec 5 2025 – Jan 14 2026
41 days · 600 positions · 3,365 position-day observations · 597 exits

Treatment variable — concentration deviation:

$$\boxed{\Delta^+ = \max\left(0,\ A_T - A_T^{1/N}\right)}$$

- $A_T = \left(\sum_k \theta_k \cdot x_k^2\right)^{1/2}$ fee concentration index (x_k = fee share, $\theta_k = 1/\text{blocklife}$)
- $A_T^{1/N} = \sqrt{\Theta/N^2}$ Ma–Crapis competitive null (equal shares $x_k = 1/N$)
- $\Delta^+ > 0$: fees more concentrated than the symmetric equilibrium

Model: The Shelter Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1 \boxed{\Delta_{t-\ell}} + \beta_2\, \Delta^2_{t-\ell} + \beta_3\, \text{IL}_t + \beta_4 \log(\text{age}_{i,t})\big)$$

We expect $\beta_1 < 0$ (shelter effect):

- Mild fee concentration correlates with *high trading volume* — the pool is active, all LPs benefit from elevated fee revenue
- At low Δ, more concentration means more volume, which *lowers* exit probability
- This is **healthy competition**: sophisticated LPs attract flow that benefits everyone [Capponi et al., 2024]
- A linear-only model would conclude concentration is *protective* — missing the adverse regime entirely

Model: The Adverse Competition Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1\,\Delta_{t-\ell} + \beta_2\,\boxed{\Delta^2_{t-\ell}} + \beta_3\,\text{IL}_t + \beta_4\log(\text{age}_{i,t})\big)$$

We expect $\beta_2 > 0$ (adverse competition):

- Beyond a threshold, concentration no longer signals volume — it signals *fee extraction* by sophisticated actors
- The quadratic term Δ^2 captures this **nonlinear regime shift**: the marginal effect of concentration *flips sign* at the turning point
- Together: $\beta_1 < 0 + \beta_2 > 0 =$ **inverted-U** with turning point $\Delta^* = -\beta_1/(2\beta_2)$

Empirical Results

Lag	$\hat{\beta}_1$ (linear)	p	$\hat{\beta}_2$ (quadratic)	p	Turning point
1	$\mathbf{-23.18}$	**0.012**	$\mathbf{+129.20}$	**0.030**	**0.090**
2	-43.42	*0.016*	$+226.92$	*0.065*	*0.096*
3	$\mathbf{-32.44}$	**0.001**	$\mathbf{+205.34}$	**0.004**	**0.079**

For detailed data curation, specification tests, and robustness checks, see `research/model/main.pdf`.

What the Data Says

Passive LPs respond to concentration shocks within one to three days. When fee dispersion exceeds ∼9% above the competitive equilibrium ($\Delta^* = -\beta_1/2\beta_2 = 23.18/(2 \times 129.20) = 0.090$), the shelter effect reverses: each additional percentage point of concentration raises exit probability by approximately 2.3 percentage points ($(\beta_1 + 2\beta_2 \cdot 0.15)\,\bar{P}(1-\bar{P}) = 15.58 \times 0.146 = 2.27$). At a fee dispersion level of 15% above equilibrium, LPs in this pool are willing to pay approximately **10% above the average hourly pool fee revenue** *for protection*
($2.3pp \times 48h$ avg remaining $\approx 1.1h$ of fees, or ∼10% of hourly revenue).

This establishes both the **existence of demand** (the inverted-U with a statistically significant adverse regime) and a **concrete price** (the marginal effect translated to an insurance premium).

Answers

Question	Finding
Is there demand for hedging adverse competition?	**Yes.** LPs react to concentration shocks within days. Below a threshold, concentration is protective; above it, adverse competition drives exits with high statistical significance.
What is the price?	LPs are willing to pay approximately **10% above the average hourly pool fee revenue** for protection — derived from the exit hazard marginal effect at 15% fee dispersion above equilibrium.

FCI Algorithm: Position Tracking

AfterAddLiquidity(*sender*, *key*, *params*)

Register position k with tick range $[a_k, b_k]$.

Snapshot cumulative fee growth at entry as baseline $F_k(0)$.

$N \leftarrow N + 1$

BeforeSwap / AfterSwap(*key*)

For every registered position k with $[a_k, b_k] \cap [p_{\min}, p_{\max}] \neq \emptyset$:

mark k as *active* (exposed to fee-generating volume).

The oracle observes two primitives — **entry** and **trading activity** — and maintains a registry of which positions were exposed to fee-generating swaps. A thin protocol adapter translates pool events into these primitives for any AMM.

FCI Algorithm: Index Accumulation

BeforeRemoveLiquidity / AfterRemoveLiquidity(*sender*, *key*, *params*)

Skip if partial withdrawal (position still active).

Compute:

Fee share (proxy): $x_k = \dfrac{L_k}{\sum_j L_j}$ — *liquidity share ≈ fee share*

Turnover: $\theta_k = \dfrac{1}{\ell_k}$ — *how fast the position turned over*

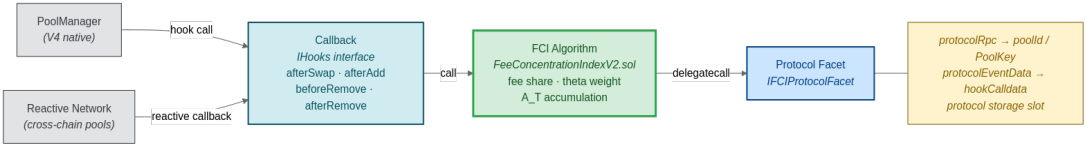
Accumulate: $A_T^2 \mathrel{+}= \theta_k \cdot x_k^2$ — *one storage slot*

$N \leftarrow N - 1$

$$\textbf{Query:} \quad \Delta^+ = \max\left(0,\ \sqrt{A_T^2} - A_T^{1/N}\right)$$

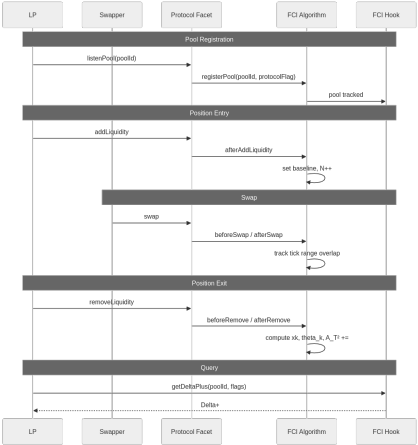
High A_T: a few short-lived positions captured most fee revenue. $\Delta^+ > 0$: concentration exceeds the competitive equilibrium. The index updates on every exit — the state is a single accumulated scalar.

Architecture



- **FCI Algorithm** warehouses the logic; facets adapt protocol data via `delegatecall`
- Facet maps: `poolRpc` → `poolId/PoolKey`, `protocolEventData` → `hookCalldata`
- Reactive Network connects pools **on demand** — not limited to V4-native (ref: `protocols/uniswap-v3/`)

Pool Listening Flow



`listenPool()` $\rightarrow$ position entry $\rightarrow$ swaps tracked $\rightarrow$ exit accumulates A_T $\rightarrow$ query Δ^+.

Metrics Interface

What an LP or indexer can query via `FeeConcentrationIndexV2`:

Function	**Returns**
`getIndex(key, flags)`	A_T, Θ, removed position count
`getDeltaPlus(key, flags)`	$\Delta^+ = \max(0, \sqrt{A_T^2} - A_T^{1/N})$
`getAtNull(key, flags)`	$A_T^{1/N} = \sqrt{\Theta/N^2}$
`getThetaSum(key, flags)`	$\Theta = \sum_k \theta_k$
`getDeltaPlusEpoch(key, flags)`	Δ^+ for current epoch window
`getRegistryActiveRanges(key, flags)`	all tracked tick ranges
`getRegistryAllSnapshots(key, flags)`	range-level snapshots (liq, swaps, fees)
`getRegistryPositionBaseline(key, flags, posKey)`	fee growth baseline at entry

Note: Epochs are essential — the cumulative A_T has infinite memory and grows monotonically. Without epoch windowing, the index reflects the entire pool history rather than recent concentration. `getDeltaPlusEpoch()` provides the time-bounded metric.

Interface: `interfaces/IFCIProtocolFacet.sol`

Reference implementation: `protocols/uniswap-v4/NativeUniswapV4Facet.sol`

Live Demo

See it by yourself

```
make sol-test-demo
```

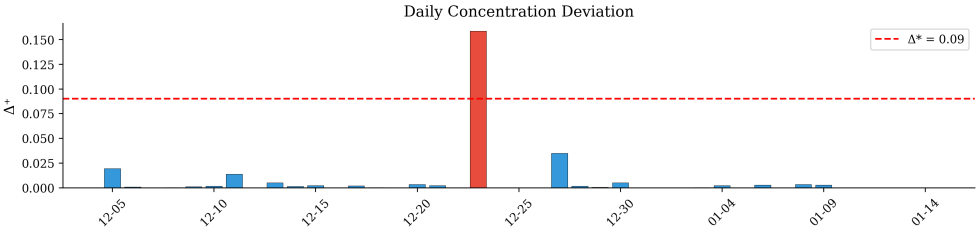
This is a **differential test**: scenarios are generated in Python, expected metric values are computed off-chain, then contrasted against the implied pool state on a Sepolia fork via `forge test`.

What to observe:

1. **Equilibrium scenario** — symmetric LPs, equal fee shares $\Rightarrow \Delta^+ \approx 0$
2. **Crowding-out scenario** — JIT-style position captures disproportionate fees $\Rightarrow \Delta^+$ spikes above Δ^*
3. **Mixed scenario** — combination of passive and sophisticated LPs $\Rightarrow$ shelter regime vs adverse regime in the same pool
4. All metrics computed on-chain, verified against Python-generated expected values

Note: NativeV4 coverage runs on Sepolia fork. 2 tests currently fail due to fixed-point precision cosmetics (rounding in x_k^2 accumulation) — to be resolved.

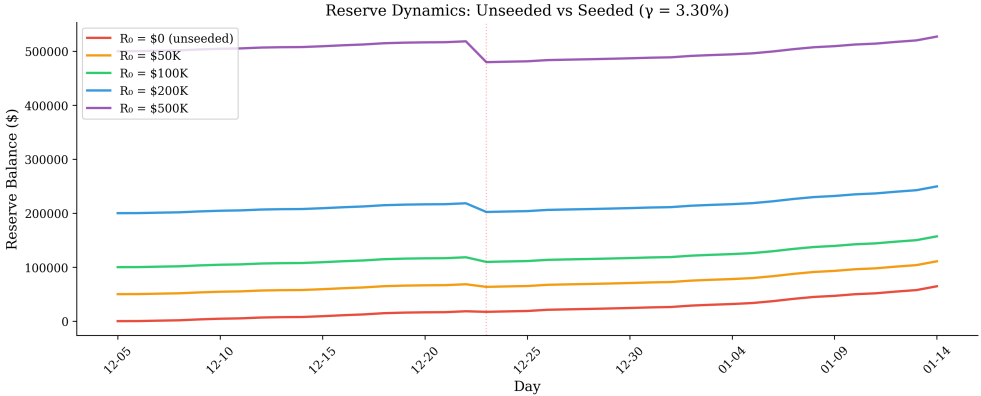
Tail Risk Confirmation



Only **1 day in 41** exceeds the turning point Δ^*. On that day, Δ^+ jumped from near-zero to **0.158**. **Rare but severe** — a classic insurable tail event.

See `notebooks/eth-usdc-backtest.ipynb` and `model/main.pdf` §5.1 for daily deviation analysis.

Backtest: Reserve Dynamics



At $R_0 = \$200\text{K}$ seed capital, the reserve absorbs the concentration spike and delivers positive hedge value.

See `notebooks/eth-usdc-backtest.ipynb` and `model/main.pdf` §3 for full methodology.

Does Hedging Improve Outcomes?

Historical replay of 600 real positions through a power-squared exit payoff ($\alpha = 2$) with \$200K underwriter seed capital:

Metric	Result
Positions better off hedged	**18.8%**
Mean hedge value	**+\$23.21** per position
Payout / Premium ratio	121.1% (payouts exceed premiums)
Optimal tail exponent	$\alpha = 2$ (dominates $\alpha = 1, 1.5, 3$)

The power-squared payoff rewards LPs who *stay in the pool* during concentration spikes — no moral hazard.

What's Next

Built

- FCI oracle live on native Uniswap V4 pools
- Reactive adapter bridging V3 events cross-chain
- Backtest validation: 18.8% of positions better off hedged, +$23 mean hedge value
- Vault prototype

Next

- Rigorous test coverage on V3 reactive integration
- Extend oracle to additional protocols
- Widen research to more pools and determine optimal sample size to strengthen the insurance hypothesis
- Optimized power-squared CFMM for insurance market making (trading function complete in spec)
- Vault settlement mechanism

The oracle tracks metrics across protocols, the backtest validates the insurance economics, and the next milestone delivers the full product.

References I

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